

Xinyu Huang

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Education

University of California, San Diego (UCSD) | Electrical and Computer Engineering, Jacobs School of Engineering | *Master of Science* 2026.09 — 2027.12 (Expected)

The University of Sydney | Software Engineering | *B.Eng (Hons I)*, GPA: Top 10% 2022.09 — 2026.06

- **Research Focus:** Security of Large Language Models (LLMs) and Embodied Agents. Published as first author on safety threat analysis of LLM-controlled robotic systems (Submitted to IEEE Transactions on Robotics, [Link: Trust in LLM-controlled Robotics: a Survey of Security Threats, Defenses and Challenges](#)).

Technical Skills²

- **Languages:** Java (Collections, JVM, GC), JUC (CAS/AQS), Python
- **Frameworks:** Spring Boot (IoC/AOP), Spring Cloud, MyBatis, FastAPI, Django
- **Databases:** Redis (Clustering, Caching), MySQL (Indexing, Transactions), Milvus, PostgreSQL
- **AI Development:** LLM, RAG, MCP, Vector DB, LangChain, ReAct, Multi-Agent, Skill
- **Else:** TCP/IP, HTTPS, OSI Model, Docker, AWS

Work Experience

Shanghai MonmentsAI Technology Co., Ltd. | *Full-stack AI Agent Development Intern* 2026.03 — Present

- Participated in the core development of a “Question Generation AI Agent,” building an end-to-end workflow based on **LangGraph** for document parsing, knowledge extraction, question generation, and quality evaluation.
- Designed a semantic splitting scheme for long documents: implemented structural segmentation with sliding window optimization for chunking granularity, combined with asynchronous metadata extraction.
- Integrated **MCP** routing with intent recognition to achieve real-time fetching and dynamic updates of exam trends; designed a **Redis caching mechanism** for MCP server queries, significantly reducing external tool latency and model hallucinations.
- Authored Dockerfiles to deploy the AgentCore module on **AWS**, effectively isolating AI inference logic from the main business system’s resources.

Project Experience

JavaQuiz | Technical Interview Prep Platform | *Backend Developer* 2026.02 — 2026.03

- **Project Background:** A Q&A practice platform for Java interviews, integrating Doubao AI APIs to provide professional feedback on user answers.
- **Architecture:** Decoupled the platform into lightweight microservices (Platform Service / AI Evaluation Service), using **RocketMQ** for asynchronous communication.
- **Authentication:** Integrated Alibaba Cloud SMS for registration/login; implemented JWT-based tokens with **Redis** for token caching and blacklist management to secure API access.
- **Context Management:** Utilized **ThreadLocal** for user session management to ensure thread isolation.
- **Cache Design:** Implemented Redis caching for hot question banks and distributed locks, reducing database load and enhancing system concurrency.

Ecommerce | Microservice-based E-commerce System | *Backend Developer* 2025.10 — 2025.11

- **Architecture:** Designed a 6-service microservice architecture (Gateway, User, Store, Order, Bank, Email) using **RocketMQ** for decoupled service communication (payment, inventory rollback, logistics).
- **High Concurrency:** Solved “overselling” issues in flash sales using **Redis + Lua scripts** for atomic inventory reduction; implemented “one order per person” logic and automatic inventory release upon payment failure.
- **Consistency & Idempotency:** Ensured interface idempotency via Redis **SETNX**; managed business flow via an Order-Payment-Inventory State Machine; utilized **delayed double-deletion** to maintain strong consistency between DB and cache.
- **Cache Optimization:** Built a multi-level cache (Caffeine local cache + Redis distributed cache); mitigated cache penetration (Bloom Filter), breakdown (mutex locks + logical expiration), and avalanche (randomized TTL).

¹ Underlined content contains hyperlinks. ² Skills irrelevant to the target position are omitted or greyed out.